

Research Assignment: Introduction to Data Science and Machine Learning

1. Define Data Science and Machine Learning. What is the relationship between them? Use a real-life example to illustrate how they work together.

- **Data Science** is the study of data and the process of deriving meaning from it and it extracts insights from structured and structured data.
- **Machine learning:** as the name indicates is a machine that learns from data fed to build algorithms and models that can be used for predictions and made decisions.
- **R/Ship:** Machine Learning is a part AI and tool of data science that uses data delivered by data science to build its algorithms and learn from it.

2. Describe the Data Science lifecycle (from problem definition to deployment). At which stage does Machine Learning typically fit in, and why?

Stage 1. Define the Problem:

- Be aware of the issue or problem you wish to resolve or eradicate.

Stage 2. Gathering Data:

- Collecting relevant data from different sources

Stage 3. Data Preprocessing:

- To clean the data by fixing missing values, duplicate records and wrong formats

Stage 4. Exploratory Data Analysis:

- Visualize the data, find patterns and identify the relationship

Stage 5. Feature Engineering:

- Create new features and normalize/scale values

Stage 6. Model Building:

- Now apply machine learning. Choose algorithm and train the model

Stage 7. Model Evaluation:

- Check if your model actually good: accuracy precision/recall

Stage 8. Deployment:

- Put the model into real use: web app, mobile app or API

Stage 9. Monitoring and Maintenance:

- Monitor performance, retrain the model with new data and fix issues.

3. Compare Supervised Learning and Unsupervised Learning, giving an example of each.

- **Supervised Learning** – The model learns from **labeled** Data, you give input + correct output and the model learns the relationships.

***Example:** predict if a patient has a chronic kidney disease*

- Input: age, blood pressure, sugar level
- Output: **Yes / No (disease or not)**

- **Unsupervised Learning** – There are no labels this model, you give only input data and the model learns hidden patterns from this data.

***Example:** Customer Segmentation*

- Input: Customer buying behavior
- Output: **No predefined labels.**

4. What causes Overfitting? How can it be prevented?

- **Causes** – Model learns too much details, Too complex Model, Small dataset, Training too long, Noisy data etc
- **Prevent** – Use more data, use train/test split and simplify the model

5. Explain how training data and test data are split, and why this process is necessary.

- **Split** – This is where you should divide your dataset into two parts

- a) **Training Dataset** – Data that the model would learn from or trained.
 - b) **Testing Dataset** – Data will be evaluated that the will understand new fed data or patterns.
 - **Why Necessary** – To check performance because if you train and test the model on same dataset it may memorize instead of learning patterns and test data acts like the future data that the model hasn't seen yet. Also to detect overfitting and gain trust that your model is working perfectly.
- 6.** Find one **case study** (research paper or article) that explains how Data Science or Machine Learning has been applied in healthcare, business, or transportation. Summarize its findings and identify which part of the Data Science lifecycle it covers.
- **Case Study** - Predicting Chronic Kidney Disease (CKD) Using Machine Learning.
 - **Research Paper** – Investigation on Explainable Machine Learning Models to Predict Chronic Kidney Diseases.
 - **Key Findings** – Machine learning models were effective in predicting CKD.
 - **DS Life Cycle** – the first five stages have been used for this research.

References:

- https://www.nature.com/articles/s41598-024-54375-4?utm_source
- <https://www.coursera.org/articles/data-science-vs-machine-learning>
- <https://www.ibm.com/think/topics/data-science-vs-machine-learning>